

Tour — Communication-Efficient Learning of Deep Networks from Decentralized Data

McMahan et al., 2017

Study repo: [pleyva2004/communication-efficient-learning-of-deep-networks](https://github.com/pleyva2004/communication-efficient-learning-of-deep-networks)

A self-contained walking tour of this study. Read top-to-bottom for a complete first pass; jump by section if you already know the foundations. Estimated total time: **90 minutes**.

1. Reader's contract

By the end of this tour you will be able to:

- Derive why FedAvg cuts communication rounds and where non-IID data breaks it
- Reproduce the headline tradeoff and the Figure-1 barrier from runnable witnesses
- Map each proposed extension to the exact gap it closes (drift, bias, plateau, shared-init)

This tour assumes comfort with multivariable calculus / gradients (refresh: [link](#)) and basic probability / expectation (refresh: [link](#)).

Out of scope: differential privacy / secure aggregation; convergence theory for $E > 1$.

2. Foundations walk

Each row is one foundation node. Links resolve to the canonical concept page in the shared [math-foundations](#) repo.

Concept	Why it matters here	Link
expectation	defines the IID identity $E[F_k] =$	link
empirical risk	<i>fundamental unbiasedness arguments</i> the finite-sum objective f is the empirical risk being minimized	link
partition of a set	the client decomposition $f = \sum_k (n_k/n) F_k$ is exact because P_k partitions the data	link
unbiased estimator	IID means each F_k is an unbiased proxy for f ; Horvitz – Thompson restores it undersampling	link
gradient	FedSGD aggregates per-client gradients into $\text{grad } f$	link
gradient descent	FedSGD with $C=1$ is exactly full-batch GD	link
taylor theorem	the $E>1$ heterogeneity gap is a second-order Taylor remainder	link

Concept	Why it matters here	Link
hessian	the gap is a covariance of local Hessians H_k with gradients	link
covariance	the drift term is exactly $\text{Cov}_k(H_k, g_k)$	link
convex function	Jensen safety of averaging needs convex F_k	link
jensens inequality	gives $f(\text{avg}) \leq \text{avg } f$ for convex objectives	link
permutation group	hidden-unit relabelings are the symmetry behind the Fig-1 barrier	link
control variates	the SCAFFOLD fix subtracts a control variate to cancel drift	link
horvitz thompson estimator	the unbiased aggregation fix is inverse-probability weighting	link
linear assignment problem	permutation alignment solves an assignment problem	link
low rank factorization	federated LoRA communicates a low-rank adapter only	link

3. Paper concepts walk

The paper graph. Read in topological order; each row links to a Markdown concept page (prose plus formal definition) and a Python witness (CPU-runnable, under 30 seconds).

Concept	Role in the paper	Files
01. Finite-Sum Objective	the objective everything minimizes	01-finite-sum-objective
02. Client-Partition Decomposition	rewrites the objective per client (exact)	02-client-partition-decomposition
03. IID vs Non-IID	defines the regime FedAvg must survive	03-iid-noniid-dichotomy
04. FedSGD as Exact Gradient Descent	the FedSGD baseline = exact GD	04-fedsgd-gradient-descent
05. Gradient- vs Model-Averaging Equivalence	why model-averaging = gradient-averaging (1 step)	05-gradient-model-averaging-equivalence
06. FedAvg: Iterated Local Steps	the FedAvg algorithm itself	06-fedavg-local-iteration
07. Local-Update Count $u = nE/(KB)$	the (C,E,B) compute-for-communication knob	07-local-update-count
08. Heterogeneity Gap = $\eta^2 \text{Cov}_k(H_k, g_k)$	the leading $E > 1$ deviation (client drift)	08-heterogeneity-gap-covariance
09. Client-Sampling Unbiasedness (Horvitz-Thompson)	justifies partial participation $C < 1$	09-client-sampling-unbiasedness

Concept	Role in the paper	Files
10. Aggregation Erratum: normalize by m_t	the published aggregation erratum	10-aggregation-erratum
11. Parameter Averaging	Jensen	why averaging is safe when convex 11-parameter-averaging-jensen
12. Shared-Init	the Permutation Barrier	why shared init is load-bearing 12-shared-init-permutation-barrier

How the pieces fit. Start from the finite-sum objective (01) and its exact client decomposition (02); the IID/non-IID dichotomy (03) says how faithful each client’s slice is. FedSGD (04) is exact GD, and the one-step gradient/model-averaging equivalence (05) is the hinge FedAvg (06) generalizes by iterating local steps — quantified by the update count (07). The two deep facts are the heterogeneity gap (08), which is exactly why non-IID hurts, and the sampling/erratum pair (09,10), which fix the aggregation. Finally (11,12) explain when averaging parameters is safe and why a shared per-round initialization is non-negotiable.

4. Improvements walk

Each proposal in `05-improvements.tex` gets exactly one validation file:

- **PROOF mode** — a standalone LaTeX proof under `proofs/` with theorem statement, complete proof, and discussion of where it would break.
- **MEASUREMENT mode** — an extended Python prototype under `improvements/` exposing `measure()` -> dict that prints a quantitative comparison table; deterministic under a fixed seed.
- **Control-Variate Drift Correction** — PROOF — `proofs/heterogeneity-control-variate.tex`
- **Unbiased Aggregation Estimators** — PROOF — `proofs/unbiased-aggregation.tex`
- **Adaptive Local-Work Schedule** — MEASUREMENT — `improvements/adaptive-local-work.py` -> `measure()`
- **Compute-Accounting Pareto (design)** — link-only — design (E.2); see `05-improvements.tex`
- **Permutation-Aligned Averaging (Git Re-Basin for FL)** — PROOF — `proofs/permutation-aligned-av`
- **Federated LoRA (design)** — link-only — design (T.2); see `05-improvements.tex`

5. What to do next

You have finished the tour. Pick one:

1. **Reproduce the sandbox claim.** `cd sandbox && pip install -r requirements.txt && python experiment.py.`

2. **Run a measurement.** Pick any improvement marked MEASUREMENT, run `python improvements/<slug>`, change one knob, re-run.
3. **Verify a proof.** Pick any improvement marked PROOF, audit the proof in `proofs/<slug>.tex`, then try to break it via the Discussion section.
4. **Form an opinion.** Fill in `03-opinions.md`.
5. **Extend the foundations graph.** Draft the missing concept at [pleyva2004/math-foundations](https://pleyva2004.github.io/math-foundations/).